POSTGRESQL + DRIZZLE ORM TO'LIQ SXEMASI

AssetAudit (MulkNazorat) — 24 ta Jadvaldan

250 ta Universitet, 10–50 milliard aktivlar, HEMIS va UzASBO/1C integratsiyasi, buxgalteriya harakatlari, modulli relyatsion arxitektura.

1-Modul: Multi-Tenant & Shartnomalar Boshqaruvi

organizations

PK: id

250 ta universitet reestri, STIR (INN), qisqa prefiks va faollik holati.

id	UUID
slug / code	VARCHAR(10) (TATU, OZMU)
name	VARCHAR(255)
status	ENUM ('active', 'frozen')
contract_end_at	TIMESTAMP

subscriptions

Shartnoma muddatlari, obuna paketi, aktivlar limiti va to'lov hisoblari.

id
organization_id
plan_tier
max_assets_limit
is_paid

2-Modul: Foydalanuvchilar, 5 ta Rol & O'zgarmas Xavfsizlik Jurnali

profiles

PK: id

5 ta rol egalari (Superadmin, Org Admin, Boshliq, Registrar, Inspector).

id	UUID
organization_id	UUID (Nullable)
full_name	VARCHAR(150)
role	ENUM (5 ta rol)
password_hash	VARCHAR(255)

user_sessions

PK: id

Auditorlar va xodimlarning faol JWT sessiyalari va qurilma ma'lumotlari.

id	UUID
user_id	UUID FK -> profile
refresh_token_hash	VARCHAR(255)
ip_address	INET
expires_at	TIMESTAMP

system_audit_logs

O'zgarmas qora quti: kim, uchun).

id
organization_id
user_id
action
old_data / new_data

3-Modul: OTM Infratuzilmasi, Bino, Qavat, Xonalar & MJS

departments

PK: id

Fakultetlar va kafedralar (Buxgalteriya va dekanatlar mas'ulligi).

id	UUID
organization_id	UUID FK -> org
name	VARCHAR(150)
type	ENUM ('faculty', 'department', 'unit')
dean_or_head	VARCHAR(150)

buildings

Binolar (Bosh bino, Yotoqxona, Sport zal, Laboratoriya, Kutubxona).

id
organization_id
code
name
type ENUM ('ac

4-Modul: Asosiy Aktivlar, Buxgalteriya Harakatlari & QR Tokenlar (YURAGI!)

funding_sources

PK: id

Mablag' manbasi: Davlat budjeti, To'lov-kontrakt jamg'armasi, Xalqaro grant, Homiylik.

id	UUID
organization_id	UUID
name	VARCHAR(100) (Budjet, Kontrakt)
code	VARCHAR(50)

suppliers

Yetkazib beruvchi firmalar (kontragentlar), shartnoma raqami va kafolat muddati.

id
organization_id
company_name
contract_number
warranty_months

assets (Bosh Jadval)

PK: id

asset_transfers

Asosiy jihoz: Inson kodi + Maxfiy Base62 token. Barcha xususiyatlar.

id	UUID
custom_label	VARCHAR(100) (E-210-001)
secret_token	UNIQUE Base62 (tk_TATU_...)
room_id	UUID FK -> room
price / initial_cost	DECIMAL(15, 2)

Xonadan-xonaga ko'chirish tarixi: 210 dan 305 ga kim ko'chirdi, kim ruxsat berdi?

id
asset_id
from_room_id
to_room_id
transferred_by / date

5-Modul: Servis, Texnik Ta'mirlash & Profilaktika

asset_repairs

PK: id

Nuqsonlarni tuzatish jurnali: usta, sarflangan pul, ehtiyot qism va kafolat hisobi.

id	UUID
asset_id	UUID FK -> asset
defect_description	TEXT (Proyektor lampasi kuygan)
repair_cost	DECIMAL(12, 2)
status	ENUM ('pending', 'in_progress', 'completed')

maintenance_schedu

Konditsioner, lift, laborator

id
asset_id
interval_months
next_service_date
responsible_technicia

6-Modul: Dala Auditi, Rektor Buyrug'i, Komissiya & Begona Buyumlar

audit_commissions

PK: id

Rektor buyrug'idagi rasmiy komissiya a'zolari (Prorektor, Bosh buxgalter).

id	UUID
organization_id	UUID
order_number	VARCHAR(100) (Rektor Buyrug'i №45)
chairman_name	VARCHAR(150) (Komissiya Raisi)
members_list	JSONB (A'zolar F.I.SH)

audit_sessions

PK: id

Xatlov davrlari (2026-yil Kuzgi xatlov, boshlanish va tugash sanalari).

id	UUID
commission_id	UUID FK -> comm
title	VARCHAR(200)
status	ENUM ('planned', 'active', 'finished')
start_date / end_date	DATE

audit_assignments

Auditorlarga binolar va xo

id
audit_session_id
inspector_id
building_id / room_id
is_completed

7-Modul: QR-Stiker Chop Etish & Rasmiy Arxiv Hujjatlari

qr_print_batches

PK: id

Ommaviy 50-100 talab QR-stiker pechatga chiqarilgan partiyalar tarixi va ro'yxati.

id	UUID
organization_id	UUID
printed_by	UUID FK -> profile
asset_ids_list	JSONB (Pechat qilingan jihozlar)
paper_size	VARCHAR(20) (40x25mm, 50x30mm)

media_documents

Siniq parta rasmlari, INV-1

id
organization_id
document_type
file_url / s3_key
file_size_kb

Arxitektura Standarti: PostgreSQL 16+ Sharded Database • Drizzle ORM Type-Safe Modellar • 0-Overhead So'